LESSON 7.7
SUBTRACTING INTEGERS USING INTEGER CHIPS

LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

LEARNING TARGETS

- I can use integer chips to subtract integers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How do you represent integer subtraction with integer chips?
- How do you use zero pairs to determine the difference of integers?

LESSON NARRATIVE

In this lesson, students begin developing an understanding of subtracting integers using integer chips. Previously students represented and solved integer addition problems using integer chips. Students connect and extend their knowledge and skills to represent and solve integer subtraction problems using integer chips.

COMPONENT SUMMARY

7.7.1 WARM-UP *(~5 MIN.)*

Math content and practices in everyday life

7.7.2 EXPLORATION *(10-15 MIN.)*

Predict the sign of the difference of two integers. Compare with a partner and check with a calculator.

7.7.3 GUIDED INSTRUCTION *(10-15 MIN.)*

Model integer subtraction with integer chips

7.7.4 GUIDED PRACTICE *(10 MIN.)*

Practice representing and solving subtraction problems using integer chips

7.7.5 YOUR TURN! *(5-10 MIN.)*

Additional practice subtracting integers using integer chips

7.7.6 WRAP-UP *(~5 MIN.)*

Demonstrate understanding of lesson learning targets

RESOURCES

GLOSSARY

Key Terms:

- Additive inverse property of zero

Additional Terms: N/A

MATERIALS

- Integer chips
- Anchor charts
- (Optional) $1 manipulatives
- Calculators

ADDITIONAL PREPARATION

- 7.7.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle determining how many zero pairs to add to the model to be able to subtract.

THINGS TO CONSIDER

- Consider guiding students through the process of how to complete the first row in the table in 7.7.2 Exploration.
- Consider creating an anchor chart during 7.7.3 Guided Instruction with images, instructions, and examples.

LITERACY AND LANGUAGE ROUTINES

Contemplate, then Calculate

Creating a plan in problem-solving before jumping in to calculating builds fluency (MA.K12.MTR.3.1) and mitigates errors. Notice the purposeful uses of "guessing" in questions 2 and 3 of 7.7.2 Exploration so students are negotiating and discussing the relationship between positive and negative integers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

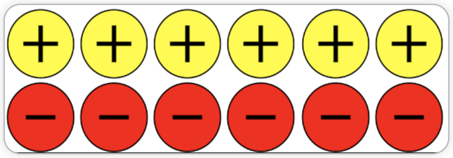
MA.K12.MTR.5.1 Patterns and Structure

7.7.4 Guided Practice provides an opportunity for students to make connections among several subtraction problems. Students develop an understanding of when, why, and how to create zero pairs and use them to subtract integers.

DIFFERENTIATE INSTRUCTION

Integer Chips

Providing concrete manipulatives offers different entry points for students to engage in mathematical discussion. Integer chips can also be printed, cut, and laminated for use in class. Consider displaying an anchor chart complete with the notes from class with images, instructions, and examples for students to reference during 7.7.4 Guided Practice and 7.7.5 Your Turn!.



7.7.1 WARM-UP: WHEN WILL I USE THIS?

Component Overview: *Students reflect on a video about the application of mathematical content and practices in everyday life. This component may help students see the utility of math. Students also reflect on the practices or thinking skills that they are developing in math.*

TEACHER MOVES

Think-Pair-Share

Give students the opportunity to share their reflections with a peer after initially reflecting independently. This will expose students to more applications of mathematics content and practices.

ENRICHMENT

Explain if you agree with the statement from the video that the math practices are more important than the math content. Provide justification for your response.

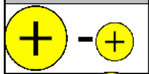
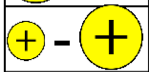
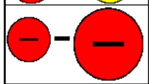
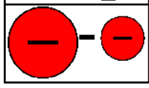
7.7.1 Warm-Up: When Will I Use This?

1. The speaker states, "It's not about what you learn, it's about what methods, tools, and tactics you have to develop in order to solve the problem that you may never see again for the rest of your life …" What does he mean by "methods, tools, and tactics"?
   Answers will vary. When the speaker says "methods, tools, and tactics" he is referring to the actions that we may take to solve a particular problem. He is stating that we may not ever have to solve a particular problem again, but the ways in which we solve that problem (the methods, tools, tactics we use) will be used over and over again in our lives.

2. The quote continues to say, "… but you will see other problems where these methods and tools will become immensely valuable to you." Where might these methods, tools, and tactics help you in everyday life?
   Answers will vary. These methods might help me in everyday life when I encounter a problem that requires me to think critically or persevere.

7.7.2 Exploration: Subtracting Integers

1. In the first column, a larger circle represents a number that has a larger absolute value than that of the smaller circle. Without discussing it with your partner, enter a numeric example for each visual model. Then, in the **"My Guess: Answer Will be + or −"** column, enter whether you think the result will be positive (+) or negative (−).
   After completing all of your own examples and guesses, record your partner's examples and guesses.
   Answers will vary. Students should have examples that match the model. For example, in line 1 the expression should have a larger positive integer subtracting a smaller positive integer.

Visual Model of Subtraction	My Example	My Guess: Answer Will be + or −	Partner's Example	Partner's Guess: Answer Will be + or −
	$8 - 4$	+	$7 - 3$	+
	$2 - 5$	−	$4 - 9$	−
	$5 - (-5)$	+	$6 - (-6)$	+
	$-5 - 5$	−	$-6 - 6$	−
	$-1 - (-7)$	+	$-2 - (-4)$	+
	$-9 - (-1)$	−	$-8 - (-2)$	−

7.7.2 EXPLORATION: SUBTRACTING INTEGERS

Component Overview: *Students predict the sign of the difference of two integers and compare their predictions with a partner. Next, students check their predictions with a calculator.*

TEACHER MOVES

Precision in size difference is less important than the discussion about the relationship between positive and negative integers, as the chips are meant to be a model or representation rather than a rote formula. Encourage students to both agree and disagree.

ENRICHMENT

- What does a larger integer chip represent in terms of absolute value?
- What happens when you subtract a larger integer from a smaller integer? A smaller integer from a larger integer?
- What happens when you subtract a negative integer?

7.7.2 EXPLORATION: SUBTRACTING INTEGERS (CONTINUED)

TEACHER MOVES

Contemplate, then Calculate

Notice the intentional scaffolding that provides students an opportunity to critically analyze and formulate a rationale (contemplate) before finding the answer (calculate). Monitor during Exploration to ensure students are actively participating in this routine for using patterns and structure to help understand and connect mathematical concepts (MA.K12.MTR.5.1).

TEACHER MOVES

Students are provided the visual model in the left column and can mark the "guess/answer" columns with dots, checks, or their preference.

It is not the expectation that all students answer these "correctly" at this point. They will continue formalizing their understanding throughout the lesson.

ENRICHMENT

- What do you notice about the sign of the difference when you subtracted an integer with a larger absolute value?
- What did you notice about the sign of the differences for each of the examples?

2. Now, form a group of four by joining another pair of classmates. Discuss your guesses. Do all four of you agree on all of the problems? If not, list which problems you disagree on.
Answers will vary.

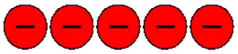
3. Tell your teacher when you are ready to test out the guesses with a calculator. Use the calculator to find the answer for one problem from each visual model. When each of you are in agreement with the rule, check the box in the column that represents the answer.

Visual Model of Subtraction	+	+	−	−
+ − +		✓		
+ − +				✓
+ − −	✓			
− − +			✓	
− − −		✓		
− − −				✓

7.7.3 Guided Instruction: Using Integer Chips to Show Subtraction

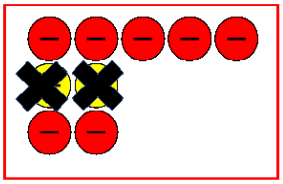
Integer chips can also be used to model subtraction.

To find the difference of $(-5) - 2$ using integer chips, begin by first arranging 5 negative integer chips on your desk, as shown below.



1. Draw a picture of how you think integer chips can be used to represent subtracting positive 2 from the 5 negative integer chips that are already on your desk.

   Consider using zero pairs. It is okay if you do not know; just try your best.
   Answers will vary and it is not the expectation students have correct models at this point.

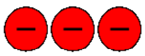
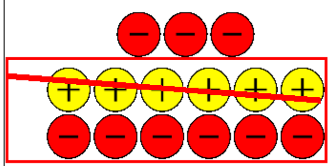


Additive Identity Property of Zero

$a + 0 = a$ or $0 + a = a$

When adding 0 to a value, the value of the expression does not change. In a model, adding zero pairs represents adding 0. The value of the model does not change, but the integer chips that are available do.

2. Complete the table to show how $(-3) - 6$ can be represented using integer chips.

Begin by representing the first integer, -3, with integer chips.	
Determine what should be subtracted from the model.	6 ○ positive ○ negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	 *Need to add 6 Zero Pairs.*
Cross out the integer chips from the model drawn above to represent the subtraction. *Shown above.*	
Find the value of the expression based on the integer chips remaining.	$-3 - 6 = -9$

3. Look at your guess for representing $(-5) - 2$. Using the directions in the table for $(-3) - 6$, determine if your guess was correct. If your guess was correct, explain how you decided you should add two zero pairs to the model.
   If your guess was not correct, explain what confused you about the zero pairs.
   Answers will vary.

7.7.3 GUIDED INSTRUCTION: USING INTEGER CHIPS TO SHOW SUBTRACTION

Component Overview: *Students model integer subtraction with integer chips. Students explore how they can use zero pairs to determine the difference.*

DIFFERENTIATE INSTRUCTION

Consider providing students with integer chips (concrete manipulatives) to provide kinesthetic and visual entry points.

ENRICHMENT

- How many zero pairs do you need to add to be able to take away 2 positive integers?
- Explain if adding zero pairs changes the value of the expression.

ENRICHMENT

How many zero pairs are needed to subtract 6 positive integers? How can you represent this using integer chips?

7.7.3 GUIDED INSTRUCTION: USING INTEGER CHIPS TO SHOW SUBTRACTION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems for students who struggle to begin writing their responses.

ENRICHMENT

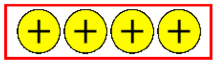
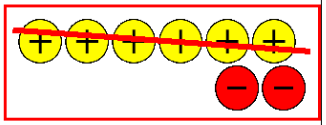
- Can you take away 6 positive integers from 4 positive integers?
- How many zero pairs need to be added to the model so that there are 6 positive integer chips to subtract from it?

TEACHER MOVES

Notice and Wonder

Ask students to share what they notice and wonder about the process of representing and subtracting integers using integer chips.

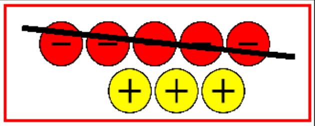
4. Complete the table to show how $4 - 6$ can be represented using integer chips.

Begin by representing the first integer with integer chips.	
Determine what should be subtracted from the model.	6 ○ positive ○ negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	 *Need to add 2 Zero Pairs.*
Cross out the integer chips from the model drawn above to represent the subtraction. *Shown above.*	
Find the value of the expression based on the integer chips remaining.	$4 - 6 = -2$

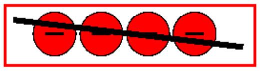
7.7.4 GUIDED PRACTICE: SUBTRACTION WITH INTEGER CHIPS

7.7.4 Guided Practice: Subtraction with Integer Chips

1. Complete the table to show how $(-2)-(-5)$ can be represented using integer chips.

Begin by representing the first integer with integer chips.	
Determine what should be subtracted from the model.	5 o positive o negative integer chips need to be subtracted from the model
If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so.	 *Need to add 3 Zero Pairs*
Cross out the integer chips from the model drawn above to represent the subtraction. *Shown above*	
Find the value of the expression based on the integer chips remaining.	$-2-(-5)=3$

2. Use integer chips to represent how $^-4 - {}^-4 = 0$.



4 negative integer chips will need to be subtracted. No zero pairs need to be added to the model because 4 negative integer chips are already available to be subtracted. After subtracting, there are no integer chips remaining, so $-4-(-4)=0$.

3. Jennifer borrowed money from Amber twice. The first time, it was \$1 for a bottle of water. The second time, it was \$5 for ice cream. Jennifer owes Amber a total of \$6. Later, Amber told Jennifer not to worry about paying her back \$2 of the \$6. The amount that Jennifer still owes Amber can be represented by the expression $-6-(-2)$.

   a. Draw integer chips to represent $-6-(-2)$ and the value of the expression.

   2 negative integer chips will need to be subtracted. No zero pairs need to be added to the model because 2 negative integer chips are already available to be subtracted. After subtracting, 4 negative integer chips remain. Therefore, the answer is -4.

   b. Determine the amount that Jennifer still owes Amber.
   Jennifer still owes Amber \$4.

Component Overview: *Students practice representing and solving integer subtraction problems using integer chips. Students then apply the skills they have developed to a real-world scenario involving money.*

ENRICHMENT

- Do you have enough negative integer chips to subtract -5? How many zero pairs need to be added to the model to have -5 available to subtract?
- Does adding zero pairs to the model change the value of the model? Does it still represent a value of -2?
- What do you think the sign of the difference will be? Why?

DIFFERENTIATE INSTRUCTION

Read the problem aloud and model withmanipulatives to provide auditory, visual, and kinesthetic entry points.

ENRICHMENT

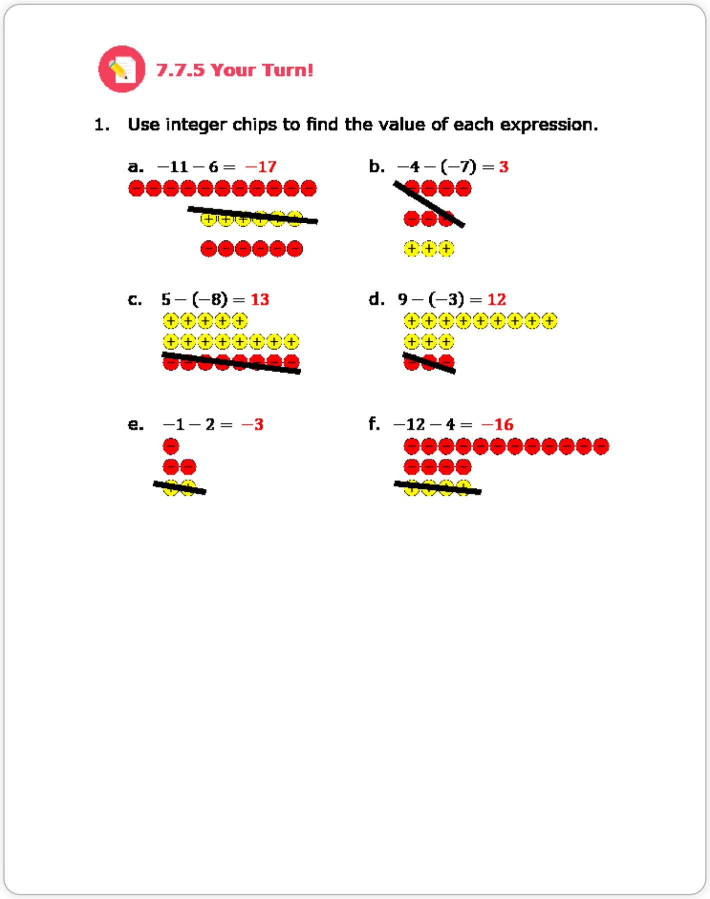
- How can you represent the original amount Jennifer borrows using integer chips?
- Will you need to use positive or negative integers? How do you know?
- Explain if you need to add zero pairs to complete the subtraction.

7.7.5 YOUR TURN!

Component Overview: *Students continue to practice subtracting signed integers using integer chips. Give students the choice of either using physical integer chip manipulatives or drawing pictorial representations of integer chips.*

DIFFERENTIATE INSTRUCTION

Consider displaying an anchor chart that displays work from 7.7.2 Exploration with images, instructions, and examples for students to reference.



7.7.5 Your Turn!

1. Use integer chips to find the value of each expression.

a. $-11 - 6 = -17$

b. $-4 - (-7) = 3$

c. $5 - (-8) = 13$

d. $9 - (-3) = 12$

e. $-1 - 2 = -3$

f. $-12 - 4 = -16$

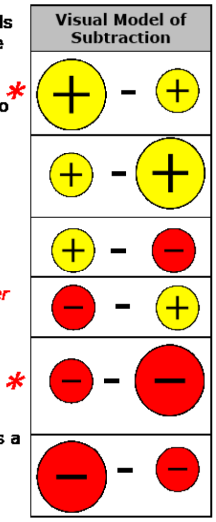
7.7.6 Wrap-Up: Cool Down

1. Choose two of the visual models of subtraction that you are sure will result in a positive answer.

   a. Draw an asterisk (*) next to both in the table.
   Answers will vary. All the correct combinations have been starred.

   b. Give an example of them below.
   Answers will vary. Sample examples are below in order from top to bottom.

   $5 - 3 = 2$
   $2 - (-5) = 7$



2. Choose one visual model that is a challenge for you.

   a. Draw an exclamation point (!) next to it in the table.
   Answers will vary.

   b. Explain using an example why it is a challenge.
   Answers will vary.

7.7.6 WRAP-UP: COOL DOWN

Component Overview: *Students demonstrate their understanding of adding integers with the same and opposite signs. Use data from this formative assessment to determine which students need support before the next lesson.*

TEACHER MOVES

Possible student responses:

- The ones with different colors because I'm not sure what to do when they are different.
- The ones that are all red because the negative and minus signs confuse me.

Consider using 7.8.3 Collaboration, in the next lesson, to connect the student responses.